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اللَّهُمَّ صَلِّ عَلَى مُحَمَّدٍ وَعَلَى آلِ مُحَمَّدٍ كَمَا صَلَّيْتَ عَلَى إِبْرَاهِيمَ وَعَلَى آلِ إِبْرَاهِيمَ إِنَّكَ حَمِيدٌ مُبْدِي، اللَّهُمَّ

بَارِكْ عَلَى مُحَمَّدٍ وَعَلَى آلِ مُحَمَّدٍ كَمَا بَارَكْتَ عَلَى إِبْرَاهِيمَ وَعَلَى آلِ إِبْرَاهِيمَ إِنَّكَ حَمِيدٌ مُبْدِي۔

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#### Q#1 What is the difference between persistent and non-persistent HTTP connections?

**Answer:** In non-persistent HTTP, each request/response pair requires a separate TCP connection to be established and closed. This requires 2 RTTs per object, and total time for N objects is  $N \times 2RTT + N \times \text{Transmit time}$ . In persistent HTTP (HTTP 1.1), a single TCP connection is used to send and receive multiple HTTP requests/responses. It requires only 1 RTT per object, and total time for N objects is  $(N+1)RTT + N \times \text{Transmit time}$ .

#### Q#2 What is the purpose of checksum in transport layer?

**Answer:** Checksum is an error detection mechanism incorporated at the transport layer. It is performed at the header or the entire body by "checking the sum" both at the sender and receiver. The sender computes the checksum and includes it in the segment header. The receiver recalculates the checksum and compares it with the received value to detect errors.

#### Q#3 Define usability in the context of network modeling.

**Answer:** Usability ( $U_b$ ) is defined as the ease of use with which network users can access the network and services. It involves ergonomic and technological facilitation where networks should make users' jobs easier. Some design decisions have a negative affect on usability, such as strict security, while some choices are user friendly, like WiFi and DHCP. Usability can be expressed as  $U_b = 1/U_e$ , where  $U_e$  is the use effort.

#### Q#4 What is the difference between availability and reliability?

**Answer:** Availability refers to the percent uptime per year, month, week, day, or hour to total time in that period. Reliability is the ability of a system to complete its function, including accuracy, error rates, and stability. A system can be available but not reliable - even if a system is available, it may not perform its functions correctly. Availability measures uptime while reliability measures correctness of operation.

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**Q#5 What are the factors affecting goodput at the application layer?**

**Answer:** Factors affecting goodput include: End-to-end error rates, Protocol functions (handshaking, windows, and acknowledgements), Protocol parameters (frame size, retransmission timers), Packet per second rate of networking devices, Lost packets at networking devices, Workstation and server performance factors (disk-access speed, disk-caching size, device driver performance), Processor (CPU) performance, Memory performance (access time for real and virtual memory), Operating system inefficiencies, Application inefficiencies or bugs.

**Q#6 What is the difference between MTBF and MTTF?**

**Answer:** The difference between MTTF (Mean Time To Failure) and MTBF (Mean Time Between Failures) is the assumption that with MTTF, the system shall be repaired, while with MTBF, the system is replaced. Both are used to assess frequency and length of service outage. MTBF is the average time between failures of a repairable system, while MTTF is the average time until failure of a non-repairable system.

**Q#7 Define scalability and Metcalfe's Law.**

**Answer:** Scalability refers to the ability to grow or add factors such as number of applications, sites, users, and servers. Metcalfe's Law states that the community value of a network grows as the square of the number of its users, expressed as  $n(n-1)/2$  or  $O(n^2)$  connections between "n" nodes. It is often cited as an explanation for the rapid growth of the Internet and can be seen in network applications.

**Q#8 What is response time in network performance?**

**Answer:** Response time is the amount of time between a request for some network service and a response to the request. It is a relative phenomenon that depends on various factors including propagation delay, transmission delay, switching delay, router delay, and queuing delay. Response time can be measured at different points in the network and is critical for interactive applications.

**Q#9 Explain the concept of jitter.**

**Answer:** Jitter is the amount of time average delay varies. Voice, video, and audio are intolerant of delay variation. Jitter is quantified in two ways: Delay jitter - bounds the maximum difference in total delay of different packets, assuming the source is perfectly periodic. Used for interactive communication like voice and video teleconferencing. Rate jitter - measures the difference in packet delivery rates at various times, measuring the difference between minimal and maximal inter-arrival times. Used for real-time applications like video broadcast.

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**Q#10 What is a DoS attack and how is it implemented?**

**Answer:** A DoS (Denial of Service) attack is an attempt to make a machine or network resource unavailable to its intended users, temporarily or indefinitely. Implementation includes: Transmit a large number of packets, TCP SYN attack, Ping attack, Server crashing attack, Large computational load. The attack success probability is  $P = 1 - (1 - 1/Q)^n$  where  $n$  packets are used for attack and  $Q$  is the queue depth.

**Q#11 What are the steps involved in measuring downtime cost?**

**Answer:** The step-wise approach to measure downtime cost includes: I. Identify Business Continuity Components (People, Property, Systems, Data), II. Define What You Protect (core competencies, product, service, process, or methodology), III. Prioritize Business Functions (business functions necessary to sustain core competency and associated IT infrastructure), IV. Classify Outage Types (Branch Outage, Regional outage, Data center outage, National outage), V. Calculate cost (Frequency x Duration x Hourly Cost = Lost Profits).

**Q#12 What are the limitations of min-hop routing?**

**Answer:** A drawback of min-hop routing is that a path with fewer hops might include slow or congested links, while a slightly longer path (more hops) could have higher bandwidth or lower latency. Also, when a demand uses indirect hops to reach its destination, it results in non-utilization of direct hops between them, and other source-destination pairs would never use these resources, leading to network partition.

**Q#13 Explain the concept of Quality of Service (QoS) routing.**

**Answer:** QoS routing is to find a "good" route for a session that meets specific requirements. For a single stream session, QoS routing finds a route for the session with given bandwidth and end-to-end delay requirements. For network operators, objectives include minimization of total bandwidth consumed and maximization of the smallest spare capacity on the links of the network. Two QoS models are IntServ (simulates virtual circuit of ATM or frame relay on layer-3 with fixed QoS parameters) and DiffServ (defining common classes of service with associated queue priorities and drop precedence on a per-hop basis).

**Q#14 What is the difference between datagram networks and virtual circuit networks?**

**Answer:** In datagram networks, two packets of the same user pair can travel along different routes, and a routing decision is required for each individual packet. In virtual circuit networks, source-to-destination paths behave much like telephone circuits with performance guaranteed. Virtual circuits require call setup

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and teardown before data can flow, each packet carries a VC identifier, and every router on source-dest path maintains "state" for each passing connection with allocated resources like bandwidth and buffers.

#### Q#15 What is the difference between Go-Back-N and Selective Repeat ARQ?

**Answer:** Go-Back-N is a retransmission strategy where when an error occurs, the entire window of N packets must be retransmitted. It does not require buffering at the receiver. Selective Repeat retransmits only packets containing errors and requires buffer requirements at the receiver. In Go-Back-N, efficiency without errors is  $E = \min\{1, N \cdot DTP/S\}$ , while with errors,  $E = 1/E[X]$ . In Selective Repeat, efficiency with errors is  $E = 1 - P$ .

#### Q#16 What is the purpose of priority queuing?

**Answer:** Priority queuing is designed to provide a relatively simple method of supporting differentiated service classes and to provide respective services to interactive traffic, voice, video, and best effort. Packets are classified and placed into different priority queues and scheduled from the head of a queue only if all queues of higher priority are empty. Within each priority queue, packets are scheduled in FIFO order. Strict priority queuing ensures packets in a high-priority queue are always scheduled before packets in lower-priority queues.

#### Q#17 What is the difference between FIFO and Fair Queuing?

**Answer:** FIFO (First In First Out) queuing is the simplest queuing discipline where packets are served in the order they arrive. During congestion, FIFO benefits UDP flows over TCP flows. Fair Queuing (FQ) ensures that each flow has fair access to network resources and prevents a bursty flow from consuming undue bandwidth share. Packets are classified into flows and assigned to a queue dedicated to that flow, then serviced one packet at a time in round-robin order. Empty queues are skipped.

#### Q#18 What are the types of simulation models in OMNeT++?

**Answer:** Types of simulation models in OMNeT++ include: INET Framework (standard protocol model library for Internet stack - TCP, UDP, IPv4, IPv6, OSPF, BGP, Ethernet, PPP, 802.11), OverSim (overlay and peer-to-peer network simulation framework for Chord, Kademlia, Pastry, GIA), Veins (Inter-Vehicular Communication simulation framework for road traffic microsimulation), INETMANET (mobile ad-hoc networks framework), MIXIM (mobile wireless, fixed wireless, WSNs, BANs, VANs, ad-hoc networks), and CASTALIA (networks of low-power embedded devices).

#### Q#19 What is the difference between additive and non-additive metrics in routing?

**Answer:** Additive link metrics are summed over the links of a path to obtain the path metric. Example: end-

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to-end delay - if each link offers  $t$  units of delay, the total links  $N$  delay is  $Nt$ . Non-additive link metrics cannot be summed over the links of a path to obtain the path metric and must be aggregated through another way. Example: Bandwidth - the least available link bandwidth along the path should be  $d$  units. If no path exists, source-destination get isolated, and bandwidth measurement frequency and accuracy is a tradeoff.

**Q#20 What is the purpose of RED (Random Early Detection)?**

**Answer:** RED (Random Early Detection) is an active queue management technique that monitors average queue size and drops packets based on probabilities. If the buffer is empty, all incoming packets are accepted. As the queue grows, the probability for dropping incoming packets grows. When the buffer is full, probability equals 1 and all incoming packets are dropped. Unlike conventional tail drop which distributes buffer space unfairly among traffic flows, RED provides intelligent drop of network packets inside a buffer of NIC to reduce network congestion.

**Q#21 What is the difference between HTTP/1.0 and HTTP/1.1?**

**Answer:** HTTP/1.0 does not support persistence and requires additional mechanisms like keep-alive. HTTP/1.1 is persistent by default and provides PUT method to upload files in entity body to path specified in URL field and DELETE method to delete files specified in the URL field. HTTP/1.1 also supports sequenced messages and persistent connections that allow multiple HTTP requests/responses to be sent and received over a single TCP connection.

**Q#22 Explain the concept of Head of Line (HOL) blocking.**

**Answer:** Head of Line blocking occurs when a queued datagram at the front of a queue prevents others in the queue from moving forward. This happens when the fabric is slower than input ports combined, causing queuing at input queues. The datagram at the front of the queue blocks other datagrams behind it, even if they are destined for different output ports. This results in queuing delay and loss due to input buffer overflow.

**Q#23 What is the purpose of ARP (Address Resolution Protocol)?**

**Answer:** ARP (Address Resolution Protocol) is used to learn the MAC address for the default router. On simulation start, Ethernet autoconfiguration precedes ARP. ARP broadcasts a request message containing the IP address to find the corresponding MAC address. The ARP packet has fields including opcode, source MAC address, destination MAC address, source IP address, and destination IP address. The packet is appended with broadcast address in control info and enqueued till ARP resolution. ARP cache builds entries with soft timers.

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**Q#24 Explain the difference between switching and routing.**

**Answer:** Routing is inter-network phenomenon that is pre-forwarding and determines end-end-path through network. Switching is intra-network phenomenon that is forwarding and involves port-based MAC learning. Routing process determines the path while switching process performs actual forwarding of packets. Comparing at device level, routers perform routing while switches perform switching. Basis of comparison includes cost (all router, all switch, hybrid), isolation (traffic, domain), speed, and complexity.

**5 Marks Questions**

**Q#1 Explain the top-down approach to modeling and simulation with a phased roll-out strategy.**

**Answer:** The top-down approach to Network Modeling and Simulation (NeMS) involves a phased roll-out of the model-simulate cycle that is iterative in nature. Networks are complex to design, so one-time design of simulation is cumbersome. The approach rolls out modeling at every layer to design a network, starting from the application layer and moving down to the physical layer. At each layer, specific parameters are modeled: Application layer (Throughput, QoS), Transport layer (TCP variants), Network layer (Router behaviour), Link layer (LAN card, switch and MAC protocol), and Physical layer (Modulation/Encoding). The strategy focuses on quantifying Quality of Experience (QoE) and parameterizing QoS at each layer. Factors considered include Scalability, Availability, Performance, Security, Manageability, Usability, Adaptability, and Affordability. The user-centric aspect ensures that the assessment of overall usability is based on the performance of network devices like hubs/switches and routers/gateways.

**Q#2 Explain the concept of Five Nines availability and its impact on costs.**

**Answer:** Five Nines availability (99.999%) means only 5 minutes of downtime per year. Some enterprises may want 99.999% availability, which is considered the best-case availability. This level of availability might require triple redundancy. The shifting impact of 9s on time shows that as availability increases, downtime decreases: 99.70% allows 1577 minutes per year, 99.90% allows 526 minutes, 99.95% allows 263 minutes, 99.98% allows 105 minutes, and 99.999% allows only 5 minutes. Hardware manufacturers may provide 5 9s, but the sum is not equal to parts due to carrier and power outages, faulty software in routers and switches, unexpected and sudden increase in bandwidth or server usage, configuration problems, human errors (90% of all), security breaches, and software glitches. Achieving Five Nines availability significantly increases costs due to the need for redundant systems, backup power, sophisticated monitoring, and skilled personnel.

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**Q#3 Explain the operation and efficiency of Slotted ALOHA with probability of success.**

**Answer:** Slotted ALOHA is designed to minimize collisions through synchronization and frame size delimiting. All frames are of the same size and time is divided into equal size slots (time to transmit 1 frame). Nodes start to transmit only at slot beginning and are synchronized. If 2 or more nodes transmit in slot, all nodes detect collision. When a node obtains a fresh frame, it transmits in the next slot. If no collision, the node can send a new frame in the next slot. If collision occurs, the node retransmits the frame in each subsequent slot with probability  $p$  until success. For  $N$  nodes with many frames to send, each transmits in a slot with probability  $p$ . The probability that a given node has success in a slot is  $p(1-p)^{N-1}$ . The probability that any node has a success is  $Np(1-p)^{N-1}$ . The maximum efficiency is found by finding  $p^*$  that maximizes  $Np(1-p)^{N-1}$ . For many nodes, taking the limit as  $N$  goes to infinity, the max efficiency =  $1/e = 0.37$  or 37%. Pros include single active node can continuously transmit at full rate, highly decentralized, and simple to implement. Cons include collisions wasting slots, idle slots, and clock synchronization needed.

**Q#4 Explain the operation of CSMA/CD and Binary Exponential Backoff algorithm.**

**Answer:** CSMA/CD (Carrier Sense Multiple Access with Collision Detection) uses carrier sensing where a node listens before transmit. If the channel is sensed idle, it transmits the entire frame. If the channel is sensed busy, it defers transmission. Collision detection occurs within a short time, and colliding transmissions are aborted to reduce channel wastage. CSMA/CD has three states: Contention, Transmission, and Idle. When a collision occurs, the Binary Exponential Backoff algorithm is used: After the  $m$ th collision, the NIC chooses  $K$  at random from  $\{0, 1, 2, \dots, 2^m - 1\}$ . The NIC waits  $K \times 512$  bit times and then returns to check if the channel is idle. If idle, it starts transmission; if busy, it waits until idle. The backoff interval becomes longer with more collisions. The efficiency of CSMA/CD is given by Efficiency =  $1/(1 + 5 \times t_{\text{prop}}/t_{\text{trans}})$  where  $t_{\text{prop}}$  is the maximum propagation delay between 2 nodes in LAN and  $t_{\text{trans}}$  is the time to transmit max-size frames. Efficiency goes to 1 as  $t_{\text{prop}}$  goes to 0 or as  $t_{\text{trans}}$  goes to infinity, and goes to 0 in the opposite case.

**Q#5 Explain the operation of TCP congestion control and the relationship between rate and congestion window.**

**Answer:** TCP congestion control is an end-end control mechanism with no network assistance. The sender limits transmission based on  $\text{LastByteSent} - \text{LastByteAcked} \leq \text{CongWin}$ , where CongWin is dynamic and a function of perceived network congestion. A loss event (timeout or 3 duplicate ACKs) causes the TCP sender to reduce the rate (CongWin). Three mechanisms are used: AIMD (Additive Increase Multiplicative

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Decrease), Slow Start, and Conservative after timeout events.  $Rate = CongWin/RTT$ . In the wake of TimeOut, RFC 2581 identifies that Slow Start is activated, bringing the window down to 1 MSS, and then exponential increase starts. TCP Reno class reference defines this operation. The relationship between throughput and loss is given by  $Throughput = (1.22 * MSS)/(RTT * \sqrt{L})$  where L is the loss rate. Fast recovery and fast retransmission are used to quickly recover from lost packets without waiting for timeout.

#### Q#6 Explain the concept of Fair Queuing with its benefits and performance conditions.

**Answer:** Fair Queuing (FQ) is a scheduling discipline designed to ensure that each flow has fair access to network resources. It prevents a bursty flow from consuming undue bandwidth share and is also called per-flow or flow-based queuing. Packets are first classified into flows by the system and assigned to a queue dedicated to that flow. Queues are then serviced one packet at a time in round-robin order, with empty queues being skipped. The primary benefit of FQ is that an extremely bursty or misbehaving flow does not degrade QoS delivered to other flows. Each flow is isolated into its own queue, so if a flow attempts to consume more than its share of bandwidth, only its queue is affected. Performance conditions for allocation of a single resource amongst N users: Total resource  $\mu_{Total}$ , each user i requests  $p_i$  and receives  $\mu_i$ . No user receives more than its request. No other user satisfying condition 1 has a higher minimum allocation. The above condition remains recursively true as we remove the minimal user and reduce total resource:  $\mu_{Total} = \mu_{Total} - \mu_i$ .  $\mu_i = \min(\mu_{Fair}, p_i)$ .

#### Q#7 Explain the operation of RED (Random Early Detection) with In and Out (RIO) packets.

**Answer:** RIO (RED with In and Out) is similar to RED but with two separate probability curves. It has two classes: "In" and "Out" (of profile). The "Out" class has a lower minimum threshold, so packets are dropped from this class first. As the average queue length increases, "In" packets are also dropped. For each packet arrival, if it is an In packet, calculate the average In queue size ( $avg\_in$ ) and calculate the average total queue size ( $avg\_total$ ). If  $min\_in < avg\_in < max\_in$ , calculate probability  $P\_in$  and with probability  $P\_in$ , drop the packet. If  $max\_in < avg\_in$ , drop the packet. If it is an Out packet, if  $min\_out < avg\_total < max\_out$ , calculate probability  $P\_out$  and with probability  $P\_out$  drop the packet. If  $max\_out < avg\_total$ , drop the packet. Since best-effort is included in the "Out" class, assured traffic can starve best-effort traffic. RIO provides differentiated services by protecting In-profile traffic while dropping Out-profile traffic first during congestion.

#### Q#8 Explain the operation and efficiency of BitTorrent.

**Answer:** BitTorrent is a peer-to-peer file sharing protocol where peers are intermittently connected and

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change IP addresses. The efficiency of BitTorrent is calculated as  $\text{Efficiency} = (\text{TBitTorrent} - \text{TCSFD}) / \text{TCSFD}$ . Factors affecting efficiency include heterogeneous upload capacity, diversities of neighbor selecting mechanisms, geographical distribution of peers, downloading rates of LocalBT clients, and peer selection policy. In BitTorrent, peers download pieces of files from multiple other peers simultaneously and upload pieces they already have to other peers. This parallel downloading and uploading creates high efficiency, especially for popular content. The protocol uses a tit-for-tat strategy where peers that upload more get better download priority. The distribution time for P2P architecture is DP2P, where total upload capacity of the system as a whole is  $u_{\text{total}} = u_s + u_1 + \dots + u_N$ , and  $d_{\text{min}}$  denotes the download rate of the peer with the lowest download rate.

**Q#9 Explain the concepts of DNS health metrics including IBC, ITV, TT, DNSR, and RRQ.**

**Answer:** DNS health metrics are used to quantify the performance and health of DNS systems. Incoming Bandwidth Consumption (IBC) is the ratio between total amount of incoming data during a session over the duration of the session, measured in Mbit/s with range  $[0, \text{IBC max}]$ . Incoming Traffic Variation (ITV) for each session  $i$  is  $(\text{IBC}_i - \text{IBC}_{i-1}) / \text{length}_i$ , where  $\text{IBC}_i$  is incoming bandwidth consumption in  $i$ th session and  $\text{length}_i$  is the duration of that session. Traffic Tolerance (TT) measures the Round Trip Time (RTT) of an IP packet flowing between an end-user node and ISP's recursive resolver in seconds. DNS Requests per Seconds (DNSR) gives the total number of DNS queries in the session. Rate of Repeated Queries (RRQ) returns the number of repeated DNS queries in a session for the same name if the query is lost or not cached. These metrics help in monitoring DNS infrastructure health and identifying potential issues like DDoS attacks or misconfigurations.

**Q#10 Explain the operation and importance of Mobile IP standard.**

**Answer:** Mobile IP is standardized in RFC 3344 and provides support for mobility in IP networks. The elements include Home agents and Foreign agents. The procedures involve Agent discovery, Registration with home agent, and Indirect routing of datagrams. The system has Foreign-agent registration and Care-of-addresses. Encapsulation (packet-within-a-packet) is used to forward packets to mobile nodes. When a mobile node moves from one network to another, it obtains a care-of address from the foreign agent and registers this address with its home agent. The home agent intercepts packets destined for the mobile node's home address and tunnels them to the care-of address. This allows the mobile node to maintain ongoing connections while moving between access networks. The degree of mobility varies from low to high mobility, where users move between access networks while maintaining ongoing connections is known as High Mobility.

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**Q#11 Explain the operation and importance of ARP (Address Resolution Protocol) in INET framework.**

**Answer:** In the INET framework, ARP operates to learn MAC addresses for IP addresses. On simulation start, Ethernet autoconfiguration precedes ARP. The ARP module processes packets and maintains an ARP cache with soft timers. The ARP packet class includes fields: int opcode enum(ARPOpcode), MACAddress srcMACAddress, MACAddress destMACAddress, IPAddress srcIPAddress, IPAddress destIPAddress. The packet is appended with broadcast address in control info and enqueued till ARP resolution. ARP variants include Broadcast-unicast behaviour, Proxy ARP, Gratuitous ARP, and Reverse ARP. Performance metrics include number of broadcast attempts, number of successes, effect of network size, and multihop performance. The ARP module output shows the ARP packet structure, packet queue containing IP packets, and the contents of ARP cache with entries and soft timers.

**Q#12 Explain the factors affecting efficiency in network performance.**

**Answer:** Factors affecting efficiency include Access protocols - high number of users showing activity can reduce efficiency, with Ethernet not efficient at high collision rates. Frame size - using large frame is useful for single user on WAN links but can cause unfair treatment. Serialization delay on WAN links results in unfair treatment for real-time shorter frames enqueued in routers. If you scale capacity more slowly than throughput while holding the average response time constant, the channel efficiency (channel utilization) will increase. The average efficiency of a network  $G$  is  $E(G) = [1/(n(n-1))] * \sum (1/d(i,j))$ , where  $n$  denotes the total nodes in a network and  $d(i,j)$  denotes the shortest path between a node  $i$  and a neighboring node  $j$ . Access protocols, frame size, and serialization delay on WAN are key factors, while hard drive size is not a factor affecting efficiency.

**Q#13 Explain the difference between additive and non-additive metrics in routing with examples.**

**Answer:** Additive link metrics are summed over the links of a path to obtain the path metric. Example: end-to-end delay - if each link offers  $t$  units of delay, the total links  $N$  delay is  $Nt$ . Other examples include hop count, propagation delay, and cost. Non-additive link metrics cannot be summed over the links of a path to obtain the path metric and must be aggregated through another way. Example: Bandwidth - requires  $d$  units of bandwidth, so the least available link bandwidth along the path should be  $d$ . Other examples include packet loss probability and reliability. Implications of non-additive metrics: If no path exists, source-destination get isolated. If more than one path exists, bandwidth measurement frequency and accuracy is a tradeoff, and bandwidth measurement is not exact. The solution is to choose the path with the highest probability of having  $d$  units.

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**Q#14 Explain the operation and types of multiplexing in communication links.**

**Answer:** Multiplexing is the process of combining multiple signals into one signal over a shared medium. Typical multiplexing techniques include: Time Division Multiplexing (TDM) where each station gets a fixed length slot, Frequency Division Multiplexing (FDM) where channel spectrum is divided into frequency bands, Statistical (Packet) Multiplexing where resources are allocated based on actual demand. Capacity overhead  $\rho$  is defined as the percentage increase in the resource requirement of a practical multiplexing scheme when compared to the optimal, calculated as  $\rho = \frac{\sum(R_i(\text{optimized}) - R_i(\text{practical}))}{\sum R_i(\text{optimized})}$ . In statistical multiplexing, with each user using 100 kb/s when active 10% of the time, strict multiplexing supports 10 users while statistical multiplexing with 35 users has probability of more than 10 active at same time less than 0.0004. Multiplexing and demultiplexing at the transport layer assigns respective data to respective ports.

**Q#15 Explain the concepts of MTBF, MTTR, and their relationship to availability.**

**Answer:** MTBF (Mean Time Between Failures) is the average time between failures of a repairable system, while MTTR (Mean Time To Repair) is the average time required to repair a failed system. Availability is calculated as  $\text{Availability} = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$ . A typical MTTF value is once per 4000 hours or 166.7 days, and a typical acceptable MTTR value is one hour, giving  $\text{Availability} = \frac{4000}{4001} = 99.98\%$  availability. MTBF with MTTR helps to assess frequency and length of service outage. MTBSO is the Mean Time Between Service Outage, and MTTSO is the Mean Time To Recover from Service Outage. The difference between MTTF and MTBF is the assumption that with MTTF, the system shall be repaired, while with MTBF, the system is replaced. Availability in percentage per annum shows different uptime levels: 99.70% allows 30 minutes downtime, 99.95% allows 5 minutes downtime.

**Q#16 Explain the operation of DNS and its support in OMNeT++.**

**Answer:** DNS (Domain Name System) provides name resolution services converting domain names to IP addresses. In OMNeT++, extensions provide classes and functions to simulate DNS and MDNS traffic implementing RFC 1035. Supported DNS operations include: Name servers with recursive resolving capabilities, Authoritative servers with DNS zone configuration using master files, Caching servers without zones (only recursively resolving), DNS Cache base that can be extended with caches based on different policies, and DNS client that can query a DNS server. DNS health metrics include Incoming Bandwidth Consumption (IBC), Incoming Traffic Variation (ITV), Traffic Tolerance (TT) measuring RTT, DNS Requests per Seconds (DNSR), and Rate of Repeated Queries (RRQ). DNS client operation is a supported DNS operation for querying DNS servers.

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**Q#17 Explain the difference between Strict Priority Queuing and Rate-Controlled Priority Queuing.**

**Answer:** Strict Priority Queuing is a variant where packets in a high-priority queue are always scheduled before packets in lower-priority queues. This ensures that high-priority traffic receives immediate service but can starve lower-priority traffic during heavy high-priority load. Rate-Controlled Priority Queuing schedules high-priority queues before lower-priority queues only if the amount of traffic in the high-priority queue stays below a user-configured threshold. This prevents starvation of lower-priority traffic while still giving priority to high-priority traffic. Both variants classify packets and place them into different priority queues. Packets are scheduled from the head of a queue only if all queues of higher priority are empty. Within each priority queue, packets are scheduled in FIFO order.

**Q#18 Explain the factors affecting delay in networks with mathematical formulation.**

**Answer:** Causes of delay include: Propagation (media type and length), Transmission (serialization) - 1024 Bytes on T1 takes specific time, Switching delay (up to 5-20 microseconds for 64 Bytes frame), Router delay (look-up, router architecture, configuration, software features like NAT, IPSEC, QoS, ACL), and Queuing delay (dependent upon utilization). The formula for queue depth is  $\text{Queue depth} = \text{Utilization} / (1 - \text{Utilization})$ . As utilization increases, queue depth increases significantly. The implications of queuing delay are that when utilization approaches 100%, queue depth approaches infinity, causing excessive delays and packet loss. The relationship between utilization and queuing delay is nonlinear, with delay increasing dramatically as utilization exceeds 70-80%.

**Q#19 Explain the operation and efficiency of Go-Back-N ARQ protocol.**

**Answer:** Go-Back-N is a retransmission strategy (ARQ) where there is no need to buffer at the receiver. When an error occurs, the entire window of N packets must be retransmitted. The efficiency without errors is: If  $N > S/DTP$ ,  $E = \min\{1, NDTP/S\}$ . With errors, where  $X = \text{Number of packets sent per successful transmission}$ ,  $N = S/DTP$ , and  $TO = NDTP$ , then  $E[X] = 1/(1-P) + (X+N)P$ , and  $E = 1/E[X]$ . The strategy requires choosing N large enough to allow continuous transmission while waiting for an ACK for the first packet of the window. Go-Back-N operates on the principle that fresher is better, and the receiver does not need to buffer out-of-order packets. However, it can be inefficient on links with high error rates because many packets may be retransmitted unnecessarily.

**Q#20 Explain the concept of minimum frame size computation in Ethernet.**

**Answer:** Ethernet recommends a minimum frame size of 64 Bytes. If the data portion of a frame is less than 46 bytes, a pad field is used to fill out the frame to the minimum size. The reasons for this minimum size

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include: Data field of 0 bytes is useful when a transceiver detects a collision and truncates the current frame. Stray bits and pieces of frames appear on cable all the time, so valid frames must be distinguished from garbage. Collision detection can take as long as  $2\tau$ , so a station must not complete transmission of a short frame before the first bit has even reached the far end of the cable where it may collide. For a 10-Mbps LAN with max length of 2500 m and  $RTT = 50 \mu\text{sec}$  in the worst case, the minimum frame must take at least this long to transmit. At 10 Mbps, a bit takes 100 nsec, so 500 bits is the smallest frame guaranteed to work. With margin of safety, round up to 512 bits or 64 bytes.

#### Q#21 Explain the operation and importance of checksum at DLL vs Transport layer.

**Answer:** Checksum at the transport layer is an error detection mechanism performed at the header or the entire body by checking the sum at both sender and receiver. In TCP and UDP, checksum is computed over all fields (header + data). In IP, checksum is computed over the IP header only. In XTP, one checksum is computed over the header and another checksum computed over the entire packet. The transport layer is typically implemented in software, so error detection has to be simple and fast, making checksumming suitable. The DLL is implemented in NIC, where CRC is more robust but more computationally intensive. Checksum detects errors by dividing the M-bit data into N-bit chunks (M/N chunks), and the checksum is also N-bit, so total sums is  $M/N + 1$ . However, checksum has limitations and cannot detect certain errors like reordering of bytes, inserting zero-valued bytes, deleting zero-valued bytes, or multiple errors which sum to zero.

#### Q#22 Explain the operation of RTP (Real-time Transport Protocol) and its support in OMNeT++.

**Answer:** RTP (Real-time Transport Protocol) is a network protocol used to transmit real-time media data, such as audio and video, over IP networks. It provides end-to-end delivery services for data with real-time characteristics including provision of timing, loss detection, and correction mechanisms. RTP is widely used in streaming media, telephony, video teleconference, television service, and push-to-talk over web. RTP works by breaking the data into small packets and sending them over the network to the intended receiver. It includes a control protocol called RTCP (Real-time Transport Control Protocol) that provides feedback to the sender about quality of transmission and enables the receiver to control data flow. RTP operates at the application layer (Layer 7) of the OSI model but is often used with UDP (Layer 4). In OMNeT++, RTP support is available in INET framework with examples like `examples/rtp/unicast1/unicast1.ned` and `src/nodes/rtp/RTPHost.ned`, providing usage diagrams and statistics collection.

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**Q#23 Explain the operation and importance of load balancing in data centres.**

**Answer:** Load balancing in data centres distributes client requests across multiple hosts to optimize resource utilization and performance. Each application is associated with a publicly visible IP address, and clients send requests and receive responses. Inside the data centre, external requests are first directed to a load balancer (also called L4 switch with NAT) which distributes and balances requests to hosts. Data centres use hierarchical topology with Top of Rack (TOR) switches interconnecting hosts in racks. However, hierarchical topologies have limited host-to-host connectivity problems where 40 simultaneous flows between 40 pairs of hosts in different racks each only receive  $10 \text{ Gbps}/40 = 250 \text{ Mbps}$ . The solution is to use fully connected topology to provide higher bandwidth and better load distribution. Load balancing can be achieved through both hardware and software approaches to efficiently distribute traffic across available resources.

**Q#24 Explain the operation of MPLS (Multi-Protocol Label Switching) and its important parameters.**

**Answer:** MPLS (Multi-Protocol Label Switching) is a fast packet switching and routing technique that provides designation, routing, and switching of traffic flows through an MPLS domain. All packets are labelled before being forwarded, and the network layer header is not processed. The basic idea is to route once and switch many times, where packets with the same traffic characteristics are forwarded in the same manner along a route from an ingress node to an egress node. MPLS network components include Label Edge Routers (LERs) at the edge and Label Switching Routers (LSRs) in the core. MPLS enhanced forwarding uses labels to determine the next hop without performing IP lookups at each router. Important parameters include Link utilization, Voice jitter, End to end delay, and Traffic Received when FRR vs link failures. MPLS supports traffic engineering and QoS, with the main goal of providing these capabilities rather than just fast packet switching.

**Q#25 Explain the difference between WiFi and WiMAX technologies.**

**Answer:** WiFi (IEEE 802.11) is a Wireless LAN technology that offers mobility and increased flexibility for portable computers, operating in high frequency bands up to 11 Mbps. It has a WLAN protocol stack and suffers from the hidden terminal problem, which is addressed by the RTS/CTS mechanism where the sender sends a request to send and the receiver acknowledges as clear. WiMAX (IEEE 802.16) is a wireless MAN technology designed to provide wireless last mile/first mile deployment and end-user access as an alternative to 802.11 family, with mobility support. WiMAX uses Line-of-Sight (LOS) and tens of GHz spectrum with severe atmospheric attenuation, making it suitable for operator networks between two nodes with high bandwidth. Many base stations are deployed at elevated positions. WiFi is suitable for

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indoor/outdoor short-range communication, while WiMAX provides longer range and higher bandwidth for MAN deployment. The protocol stack for WiFi includes Logical Link Control, Radio, and 802.11 MAC, but does not include Passive Optical Network.



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